Before integration (11 pages)

After integration (13 pages)

Learning When Black-Hole Hair Is Observable: Physical Identifiability, Generalization, and Observable Complementarity

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Abstract

High interpolation accuracy does not establish that physical parameters are identifiable. We study this distinction in a controlled Kerr black-hole benchmark with a validated timelike-emitter and three-dimensional null-geodesic shooting calculation. The forward analysis shows that ringdown is informative for the deformation parameter ζ, but the observable response direction is dependent response direction that resolves much of the state-parameter ν_ζ degeneracy. Relative to ringdown alone, ringdown plus photon geometry improves the grid-median minimum singular value by 4.54× and the median Jacobian condition number by $2.29\times$. In grouped physical interpolation, the corresponding identifiable-ν_ζ normalized mean absolute error decreases from 0.309 for ringdown to 0.087 for the combined features. Random splitting is more optimistic, while four directional extrapolation tests are substantially harder and split-conformal coverage falls under distribution shift. A uniform 161-phase audit gives a typical normalized prediction shift of 9.48×10^{-3} but exposes catastrophic extrapolation tails. A subsequent targeted 321-phase audit resolves the forward-feature criterion but does not by itself certify inverse estimators. Bayesian parameter estimation using HGB and random-forest tail shifts exceed predeclared robustness thresholds. Thus the physical complementarity result is supported, but inverse estimator robustness remains conditional. This study is a theoretical identifiability benchmark, not an observational constraint.

1 Introduction

Scientific inverse models can predict accurately for the wrong reason. Repeated realizations of one physical system can look across random partitions; parameters can be locally degenerate even when observables respond strongly; and a model can interpolate well while failing under extrapolation or small numerical perturbations. These distinctions matter in black-hole spectroscopy, where quasinormal frequencies encode compact-object properties but resolving multiple physical effects remains difficult [8, 2, 4].

Kerr spacetimes provide a controlled two-parameter setting. They do not stand in for a detector likelihood or a generic beyond-Kerr theory. Their value here is methodological: exact loss of ν_ζ dependence at $\lambda = 0$ supplies a known rank-deficient boundary, while nonzero λ permits direct comparison of forward Jacobian geometry and inverse-learning behavior [12].

The study complements rapid gravitational-wave inference based on neural posterior estimation and normalizing flows [7, 11, 10, 6, 16]. Those methods address posterior computation; the present question is whether the data and the chosen observables distinguish the parameters in the first place. This distinction parallels structural and practical identifiability analysis in dynamical inverse problems [18, 24].

This work makes four contributions. It (i) evaluates inverse models with leakage-aware random, grouped, and directional protocols; (ii) validates timelike and null Kinnersley geodesic shooting with explicit failure accounting; (iii) quantifies physical Jacobian conditioning and observable complementarity; and (iv) audits uncertainty, noise, learning curves, and 81–161–321 phase-resolution robustness under frozen estimators.

The central narrative rests on three distinctions: sensitivity is not identifiability, interpolation is not generalization, and forward convergence is not estimator robustness.

2 Physical Model and Identifiability Framework

The static spherical line element is

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2 d\Omega^2, \tag{1}$$

$$f(r) = 1 - \frac{2M}{r} + \frac{\zeta}{r^3(r-\tfrac{1}{2}M)}. \tag{2}$$

We use geometrized units with $M = 1$. The dense historical synthetic domain and the latter 121-point physical shooting domain are distinct datasets and are not mixed. Leading-ringdown ringdown features use the unstable circular null orbit, with the known limitations of the geodesic–QNM correspondence kept explicit [5, 14, 23].

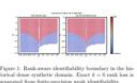


Figure 1: Rank-aware identifiability boundary in the historical dense synthetic domain. Exact $\lambda = 0$ rank loss is separated from finite-precision weak identifiability.

For standardized observables $\mathbf{O}$ and parameters $\boldsymbol{\theta} = (\lambda, w_\zeta)$, the local Jacobian is

$$J_\zeta = \frac{\partial \mathbf{O}}{\partial \theta_\zeta}. \tag{3}$$

Higher $\sigma_{\min}(J)$ and lower $\kappa(J) = \sigma_{\max}/\sigma_{\min}$ are favorable. At $\lambda = 0$, $f(r)$ is Schwarzschild for every w_ζ, so the w_ζ Jacobian column vanishes exactly. Numerics rank away from that boundary does not by itself imply practical recoverability at finite precision. Figure 1 records the rank-aware historical benchmark that motivated the matter-physical calculation.

3 Physical Geodesic Shooting and Numerical Validation

The emitter follows the equatorial timelike Hamiltonian

$$H = \tfrac{1}{2}\tilde{h}^{ij}(f + f\tilde{p}_r^2 + L^2/r^2) = -\tfrac{1}{2}. \tag{4}$$

Its physical coordinate azimuth begins at apocentric $\phi = \pi$, $t = r_a$, $p_r = t = \tau = 0$. Coordinate ϕ is not treated as a radial anomaly; turning points are detected from $p_r = 0$ and the radial azimuthal period includes spatial advance.

Photons are propagated in Cartesian spatial coordinates with covariant momentum $\mathbf{p}$ and $p_t = -1$,

$$\dot{H}_\gamma = \tfrac{1}{2}(-1/f + \mathbf{p}^2 \circ (f - 1)\,\mathbf{n}^{(r)}) = 0. \tag{5}$$

Two launch angles are solved so the direct branch reaches the observer plane. The implementation reports failed planes rather than interpolating them or substituting synthetic geodesic proxies. Photon momenta are projected onto the static observer tetrad before sky angles are formed. Redshift uses $1 + z = p_{\text{emit}}/p_{\text{obs}}$, and direct arrival time is compared with $f(1 + z)M_{\text{true}}$ after removing one common additive constant.

Across the accepted physical grid, ringdown plus photon geometry improves the median $\sigma_{\min}$ by 4.542× and the median condition number by $2.293\times$ relative to ringdown alone. The medians for the combined set are $\sigma_{\min} = 2.2336$ and $\kappa(J) = 4.8731$. Ringdown plus all shooting has $\sigma_{\min} = 1.5312$ and $\kappa(J) = 4.4974$: adding more responsive summaries can raise $\sigma_{\min}$ without producing the best conditioning. Figures 5 and 6 show the posterior maps and their grid-wide summaries, respectively.

Table 1: Primary physical-complementarity summaries. The reported combined-set gains are the archived grid-wide improvement statistics relative to ringdown, not ratios formed from separately reported medians.

Features	$\mathrm{med}(\sigma_{\min})$	$\mathrm{med}(\kappa(J))$	gain
Ringdown	0.2762	6.6112	reference
Ringdown + photon geometry	1.2538	4.6731	$4.54\times$ / $2.29\times$
Ringdown + all shooting	1.5312	4.4974	

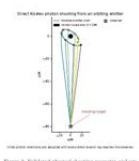


Figure 3: Validated physical shooting geometry and representative direct photon rays. The emitter begins at apocenter at $\phi = \pi$; trajectories were regenerated from archived converged launch angles without proxy substitution.

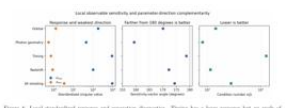


Figure 4: Local standardized response and separation diagnostics. Timing has a large response but an angle of 178.76° and condition number 1275.5; photon geometry rotates the response to 159.27° and lowers the condition number to 81.4. Higher $\sigma_{\min}$, angles farther from 180°, and lower condition number are favorable; magnitude alone is not identifiability.



Figure 5: Grid maps of $\log_{10}\sigma_{\min}$ with exact $\lambda = 0$ rank-loss points masked. Missing or invalid values are not interpolated, and all panels use one finite-4 color scale.

4 Physical Observable Complementarity

A large response can reveal that the spacetime changed without revealing which parameter caused the change. Timing is the strongest single feature, but nearly collinear in parameter space. Photon geometry rotates the sensitivity direction and therefore contributes information that ringdown lacks. The standardized local comparison is shown in Fig. 4.

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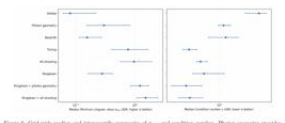


Figure 6: Grid-wide median and interquartile summaries of $\sigma_{\min}$ and condition number. Photon geometry provides the clearest complement to ringdown.

5 Leakage-Aware Inverse-Learning Protocols

Five primary feature sets are ordered consistently: ringdown, photon geometry, all shooting, ringdown plus photon geometry, and ringdown plus all shooting. Here photon geometry and the other integrated quantities are physical shooting observables, distinct from the historical synthetic geodesic proxies. Models are histogram gradient boosting (HGB), random forest (RF), and a target-scaled multilayer perceptron (MLP) [4, 9, 17]. The protocols are random interpolation, two blocked 3×3 grouped physical layouts, and four directional extrapolation tests. All nominal $\lambda = 0$ records are retained and flagged, but ordinary w_ζ error excludes them because w_ζ is exactly non-identifiable there.

Normalized mean absolute error divides MAE by the full target span. Split–conformal intervals target 90% coverage and are interpreted under their exchangeability assumptions [22, 20, 1]. Feature-noise scales and every preprocessing transform are learned on training folds. The 81-to-161 and targeted 161-to-321 comparisons apply archived frozen estimators; they do not retrain on higher-resolution test features.

6 Inverse-Learning Results

Random splitting is optimistic. For HGB, RF, and MLP considered together, random ringdown NMAE is 0.0361/0.1609 for λ/identifiable w_ζ, while grouped ringdown is 0.0405/0.3093. Ringdown plus photon geometry gives grouped 0.0457/0.0875, with the w_ζ improvement reproduced independently in both tree families. Directional extrapolation remains harder: the combined set has aggregate NMAE 0.1305/0.2012. Figure 7 compares the three validation regimes directly.

Conditioning and coverage explain different aspects of error. In the error model, standardized training distance has estimate 0.456, extrapolation status 0.276, and log condition number 0.197. Physical conditioning generally agrees with grouped tree-based w_ζ performance, but it does not perfectly rank every model, target, or protocol. The MLP supplies a counterexample: it can have reasonable central metrics while producing catastrophic directional tails. The aligned physical and model diagnostics appear in Fig. 8.

Nominal 90% conformal coverage is 0.892/0.965 for random λ/w_ζ, but falls to 0.722/0.851 under grouped interpolation and 0.508/0.666 under extrapolation. This is empirical coverage under shift, not a retained finite-sample guarantee. Figure 9 separates coverage by target and protocol.

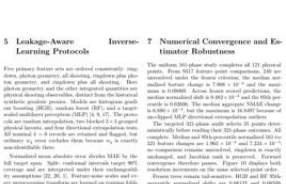


Figure 7: Median NMAE for the five primary feature sets under random, grouped physical interpolation, and directional extrapolation. Directions remain separate in the machine-readable source before aggregation.



Figure 8: Aligned physical and empirical diagnostics: median minimum singular value, median condition number, grouped identifiable-w_ζ NMAE, and grouped λ NMAE. Exact statistics and the feature order are shared across panels.

MLP is smoother under the targeted feature substitution but extrapolates to physically unreasonable values in archived directional tests. These mechanisms are estimator-specific manifestations of a general scientific-ML risk: a converged forward simulator does not ensure a stable inverse surrogate [15, 13, 19].

What is specific to Kinnersley is the metric deformation and exact $\lambda = 0$ loss of w_ζ. What may generalize is the workflow: establish forward rank, seek complementary observables, partition by physical systems, test directional shift, and perturb frozen estimators at independently measured numerical scales.

7 Numerical Convergence and Estimator Robustness

The uniform 161-phase study completes all 121 physical points. From 9317 feature-point comparisons, 240 are unresolved under the frozen criterion; the median normalized feature change is 7.908×10^{-8} and the maximum is 0.00669. Across frozen scored predictions, the median normalized shift is 9.482×10^{-3} and the 95th percentile is 0.0486. The median aggregate NMAE change is 6.000×10^{-5}, but the maximum is 16.6497 because of two-flipped MLP directional extrapolation outliers.

The targeted 321-phase audit selects 35 points deterministically before reading their 321-phase outcomes. All complete. Median and 95th-percentile normalized 161-to-321 feature changes are 1.863×10^{-4} and 7.234×10^{-3}; no comparison remains unresolved, ringdown is exactly unchanged, and Jacobian rank is preserved. Forward convergence therefore passes. Figure 10 displays both resolution increments on the same selected-point order.

Frozen trees remain tail-sensitive. HGB and RF 95th-percentile normalized shifts are 0.08122 and 0.08509, and their largest identifiable-w_ζ shifts are 0.30593 and 0.27053. Their median shifts are zero because many tree predictions do not cross a split threshold. The predeclared perturbation ensembles and fixed Extra Trees smoother do not pass all stability and grouped-performance criteria. Of 410 catastrophic MLP records, 408 are already catastrophic at 161 phases, none is materially worsened by the 321-phase substitution, 63 reach the iteration limit, and no numerical overflow is detected. The decision is Outcome B: the forward map converges, but estimator robustness does not.

Typical predictions are stable, but the tails are not. The earlier MLP mantissa are not representative of every estimator, while the targeted test failures show that the remaining issue also cannot be attributed solely to the MLP. *Formal* realness remains conditional.

Figure 11 separates the estimator mechanisms. Tree medians remain exactly unchanged because most perturbations do not cross a split, yet their tails do. The MLP reacts more smoothly to phase substitution, while its catastrophic directional records overwhelmingly predate that substitution.

8 Discussion

Jacobian conditioning answers whether nearby parameter directions are separated by the chosen observables. Training distance answers whether a fitted estimator has support where it is evaluated. Neither statistic subsumes the other. Consequently, $\sigma_{\min}$ can predict broad observable-set ordering without ranking every empirical model or protocol.

Tree ensembles are piecewise and can show zero median response followed by large split-crossing tails. The uniform 161-phase study completes all 121 physical points. From 9317 feature-point comparisons, 240 are unresolved under the frozen criterion; the median normalized feature change is 7.908×10^{-8} and the maximum is 0.00669. Across frozen scored predictions, the median normalized shift is 9.482×10^{-3} and the 95th percentile is 0.0486. The median aggregate NMAE change is 6.000×10^{-5}, but the maximum is 16.6497 because of two-flipped MLP directional extrapolation outliers.

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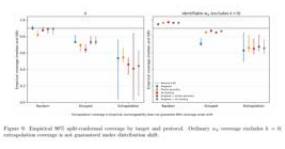


Figure 9: Empirical 90% split-conformal coverage by target and protocol. Ordinary w_ζ coverage excludes $\lambda = 0$; extrapolation coverage is not guaranteed under distribution shift.

Table 2: Targeted-321-phase frozen-estimator robustness.

Model	median	95th percentile	max. identifiable w_ζ
HGB	0	0.08122	0.30593
RF	0	0.08509	0.27053
MLP	9.063143	9.809223	9.83819

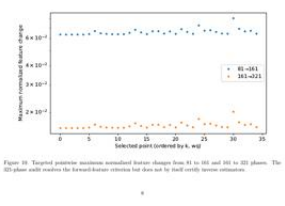


Figure 10: Targeted pointwise maximum normalized feature changes from 81 to 161 and 161 to 321 phases. The 321-phase audit resolves the forward-feature criterion but does not by itself certify inverse estimators.

9 Limitations

The ringdown model is leading-chiral rather than a distorted Hartle-ℓ perturbative spectrum. The spacetime is static. Shooting includes only the direct photon branch, a point emitter and point observer, and no radiative transfer, detector response, secondary images, strain likelihood, calibration error, or model discrepancy. The physical grid is finite. Historical proxy results remain synthetic. Although the targeted audit resolves higher-harmonic forward differences on its selected set, it is not a uniform 321-phase grid and tree tail sensitivity persists. MLP extrapolation is unstable. The verdict is therefore conditional, and no result is an observational constraint.

10 Conclusion

Physical complementarity is supported: ringdown constrains λ, while photon geometry supplies an independent direction needed for much of the w_ζ recovery. Ringdown interpolation overstates recoverability relative to grouped and directional validation. Finally, numerical and distribution-shift audits are necessary before confident inverse claims: the forward features can converge while fitted estimators remain fragile. These conclusions motivate robust inverse methods and broader physical modeling before observational use.

A Historical Synthetic and Proxy Benchmark

The original dense benchmark used leading-chiral curves, training-fold PCA, and synthetic geodesic proxies. Waveform-only RF achieved $R^2(\lambda) = 0.9960$ and $R^2(w_\zeta) = 0.9507$. HGB with proxy augmentation achieved 0.9793 and 0.9091. The rank-aware classifier reached macro F1 0.9928 with a 57.6% weak-identifiability fraction. PCA compressed correlated information; it did not create a new physical response direction. These historical results motivated the physical shooting study and are not substituted for it.

B Full Split Definitions

Random splits partition parameter points before model evaluation. Grouped physical interpolation holds out blocks under both registered 3×3 layouts. Directional extrapolation separately holds high-λ, low-λ, high-w_ζ, and low-w_ζ sides. Calibration points are disjoint from training and test points, and the archived assignments are reused exactly.

C Additional Model-Level Results

HGB and RF reproduce the central grouped photon-geometry improvement for identifiable-w_ζ. The MLP is retained as a stress test rather than omitted; its extrapolation errors are unclipped and included in aggregate summaries.

D Four Directional Extrapolation Tests

Directions are stored independently in the prediction tables before any median summary. High-to-low and low-to-high tests differ in training support and cannot be interpreted as interchangeable folds.

E Conformal Calibration and Rejection

Split-conformal residual quantiles are fitted on calibration subsets. The exchangeability guarantee applies to matching distributions; shifted coverage is reported empirically. Rejection curves rank uncertainty but do not repair distribution shift.

F Noise and Learning Curves

Grouped one-percent feature noise yields ringdown-plus-photon-geometry NMAE 0.0699/0.0853 for λ/w_ζ. Learning curves are still improving at 78 training points, so data limitation remains part of the interpretation.

G Numerical-Validation Thresholds

The 35-point 321-phase run takes 5214.2 s. Maximum fat, timelike-constraint, null-constraint, and impact-parameter drift errors are respectively 7.0913×10^{-8}, 3.1303×10^{-13}, 7.6214×10^{-13}, and 1.7922×10^{-14}. Matched Schwarzschild and Kinnersley direct rays are compared on their actual coordinate scale in Fig. 14.

H Uniform 161-Phase Convergence Audit

All 121 points complete in 4173.2 s with zero failed points. The audit preserves the exact $\lambda = 0$ boundary and uses no mixed-resolution row, interpolation, or proxy substitution.

I MLP Extrapolation-Failure Audit

The machine-readable table contains standardized input magnitude, hidden activation magnitude, iteration status, overflow flag, training distance, and resolution response for every catastrophic record. It distinguishes optimization failure from extrapolation instability without clipping.

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